F70LA Life Insurance Mathematics A (2025/26)
Tutorial 1 Solutions

1. (a) (i) $\mathbb{P}[T_0 \leq 65] = F_0(65) = \frac{65}{80}$.
   (ii) $\mathbb{P}[T_0 > 50] = 1 - F_0(50) = 1 - \frac{50}{80} = \frac{30}{80}$.
   (iii) $\mathbb{P}[40 < T_0 \leq 45] = F_0(45) - F_0(40) = \frac{45}{80} - \frac{40}{80} = \frac{5}{80}$.
   (iv) $\mathbb{P}[33 < T_0 \leq 34] = F_0(34) - F_0(33) = \frac{34}{80} - \frac{33}{80} = \frac{1}{80}$.

   (b) Denote the pdf by $f_0(t)$.

$$f_0(t) = \frac{d}{dt}F_0(t) = \begin{cases} \frac{d}{dt}\left(\frac{t}{80}\right) = \frac{1}{80} & \text{for } 0 \leq t \leq 80 \\ \frac{d}{dt}(1) = 0 & \text{for } t > 80. \end{cases}$$

   The pdf is constant at all ages less than 80 years. That means the individual has the same chance of dying in every year, up to age 80years. The individual is certain to die by age 80 years, under this mortality model.

   (c)

$$\mathbb{E}(T_0) = \int_0^\infty t \cdot f_0(t)dt = \int_0^{80} \frac{t}{80}dt = \left[\frac{t^2}{160}\right]_{t=0}^{t=80} = 40.$$

   Therefore, the expected future lifetime of the newborn is 40 years. The expectation of T_0 is also known as complete expectation of life (from age 0, in this case). In international actuarial notation, it is written for the newborn as $\mathring{e}_0 = 40$.

   (d) As

$$\mathbb{E}(T_0^2) = \int_0^\infty t^2 \cdot f_0(t)dt = \int_0^{80} \frac{t^2}{80}dt = \left[\frac{t^3}{240}\right]_{t=0}^{t=80} = \frac{80^2}{3},$$

   the variance of T_0 is

$$\text{var}(T_0) = \mathbb{E}(T_0^2) - (\mathbb{E}(T_0))^2 = \frac{80^2}{3} - 40^2 = \frac{80^2}{12}.$$

   Hence the standard deviation of T_0 is

$$\text{sd}(T_0) = \sqrt{\text{var}(T_0)} = \frac{80}{\sqrt{12}} = 23.094 \text{ years}.$$

2. Note that $S_0(t) = \mathbb{P}[T_0 > t] = e^{-(0.002t + 0.000023t^2)}$.

   (a) $\mathbb{P}[T_0 \leq 60] = F_0(60) = 1 - e^{-(0.002(60) + 0.000023(60)^2)} = 1 - e^{-0.2028} = 0.18356$.

   (b) $\mathbb{P}[T_0 > 80] = S_0(80) = e^{-(0.002(80) + 0.000023(80)^2)} = e^{-0.3072} = 0.73550$.

(c)

$$\begin{aligned}\mathbb{P}[55 < T_0 \leq 65] = F_0(65) - F_0(55) =& e^{-(0.002(55)+0.000023(55)^2)} - e^{-(0.002(65)+0.000023(65)^2)} \\ =& e^{-0.1796} - e^{-0.2272} = 0.03884.\end{aligned}$$

(d)

$$\begin{aligned}\mathbb{P}[40 < T_0 \leq 45] + \mathbb{P}[50 < T_0 \leq 55] &= (F_0(45) - F_0(40)) + (F_0(55) - F_0(50)) \\ &= \left(e^{-0.1168} - e^{-0.1366}\right) + \left(e^{-0.1575} - e^{-0.1796}\right) \\ &= 0.03607.\end{aligned}$$

3. (a) $F_0(60) = 1 - e^{-60\lambda}$. $F_0(60)$ represents the probability of a newborn dying before age 60.

   (b) As the constant λ increases, the probability of dying before age 60 years increases, e.g. $\lambda \uparrow \Rightarrow -60\lambda \downarrow \Rightarrow e^{-60\lambda} \downarrow \Rightarrow 1 - e^{-60\lambda} \uparrow$.

   (c)

$$f_0(t) = \frac{d}{dt}F_0(t) = \frac{d}{dt}\left(1 - e^{-\lambda t}\right) = \lambda e^{-\lambda t}.$$

   (d) The survival function is $S_0(x) = e^{-\lambda x}$. The force of mortality at age x is

$$\mu_x = -\frac{d}{dx}\left(\ln(S_0(x)\right) = -\frac{d}{dx}\left(-\lambda x\right) = \lambda.$$

   Equivalently, $\mu_x = f_0(x)/S_0(x) = \lambda e^{-\lambda x}/e^{-\lambda x} = \lambda$.

4. (a)

$$F_x(t) = \frac{F_0(x+t) - F_0(x)}{1 - F_0(x)} = \frac{(1 - e^{-0.01(x+t)}) - (1 - e^{-0.01x})}{1 - (1 - e^{-0.01x})} = 1 - e^{-0.01t}.$$

   To derive an expression for $S_x(t)$, either $S_x(t) = 1 - F_x(t) = e^{-0.01t}$ or, noting that $S_0(t) = e^{-0.01t}$,

$$S_x(t) = \frac{S_0(x+t)}{S_0(x)} = \frac{e^{-0.01(x+t)}}{e^{-0.01x}} = e^{-0.01t}.$$

   (b) The expressions do not depend on the age x, which is the 'starting' age of the individual. This means that a 30-year-old has the same chance of dying as a 90-year-old, which is not seen in human mortality statistics. This 'independence of age' is a feature of this 'constant force of mortality' model.

   (c) (i) $\mathbb{P}[T_{30} \leq 30] = F_{30}(30) = 1 - e^{-0.3} = 0.25918$.

   (ii) $\mathbb{P}[T_{50} > 30] = S_{50}(30) = e^{-0.3} = 0.74082$.

   (iii)

$$\mathbb{P}[15 < T_{40} \leq 25] = F_{40}(25) - F_{40}(15) = \left(1 - e^{-0.25}\right) - \left(1 - e^{-0.15}\right) = 0.08191.$$

(iv)

$$\begin{aligned}
&\mathbb{P}[10 < T_{30} \leq 15] + \mathbb{P}[20 < T_{30} \leq 25] \\
=&(F_{30}(15) - F_{30}(10)) + (F_{30}(25) - F_{30}(20)) \\
=&\left(1 - e^{-0.15}\right) - \left(1 - e^{-0.1}\right) + \left(1 - e^{-0.25}\right) - \left(1 - e^{-0.2}\right) \\
=&e^{-0.1} - e^{-0.15} + e^{-0.2} - e^{-0.25} = 0.08406.
\end{aligned}$$

5.

$$\begin{aligned}
F_x(t) &= \frac{F_0(x+t) - F_0(x)}{1 - F_0(x)} = \frac{(1 - e^{-(\alpha(x+t)+\beta(x+t)^2)}) - (1 - e^{-(\alpha x+\beta x^2)})}{1 - (1 - e^{-(\alpha x+\beta x^2)})} \\
&= \frac{-e^{-\alpha(x+t)-\beta(x+t)^2} + e^{-\alpha x-\beta x^2}}{e^{-\alpha x-\beta x^2}} \\
&= 1 - \frac{e^{-\alpha(x+t)-\beta(x^2+2xt+t^2)}}{e^{-\alpha x-\beta x^2}},
\end{aligned}$$

i.e.

$$F_x(t) = 1 - e^{-\alpha t-\beta(2xt+t^2)}.$$